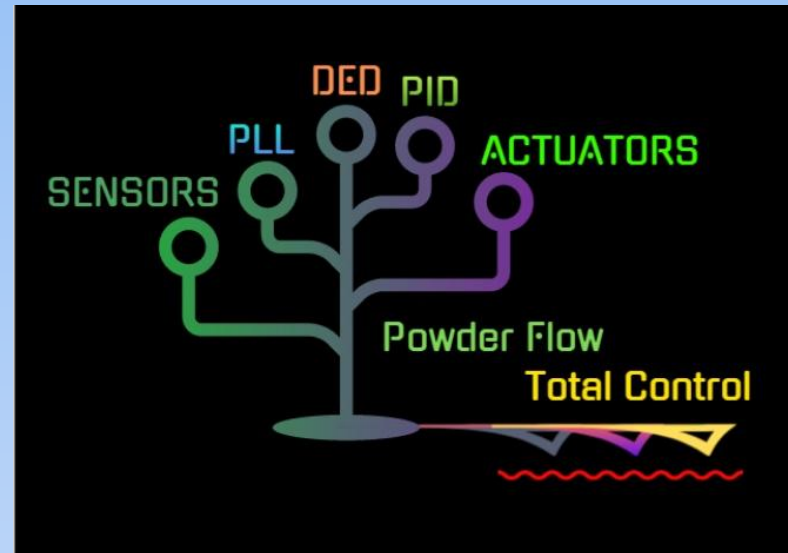
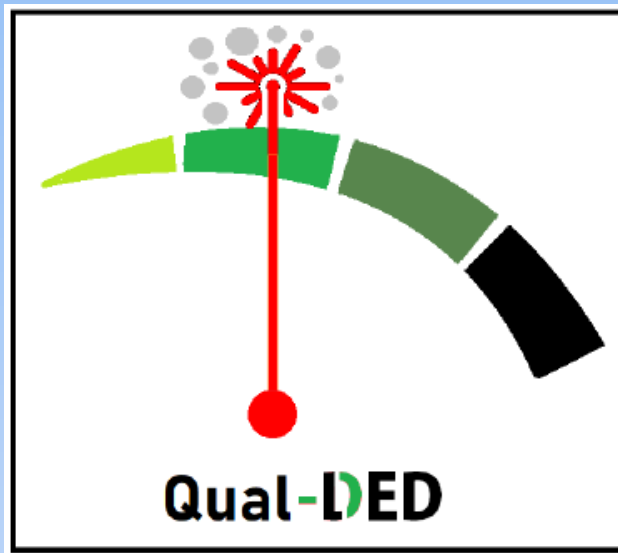




Powder Flow Total Control

Powder flow control of L-DED based additive manufacturing for zero defect metal components



Project work defence 09.07.2020
Alessandro Corgiolu

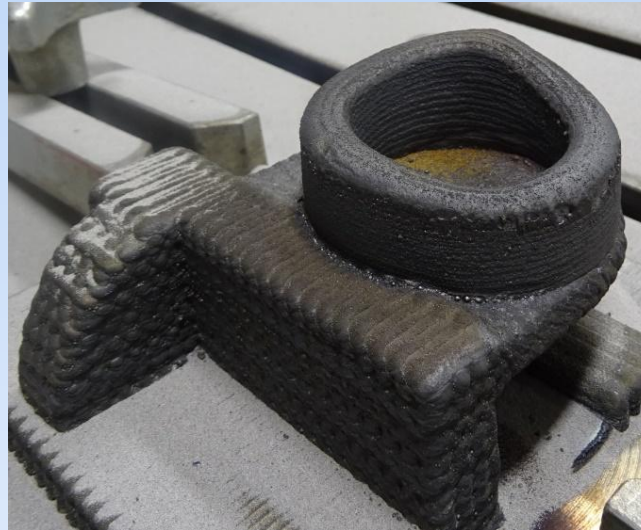
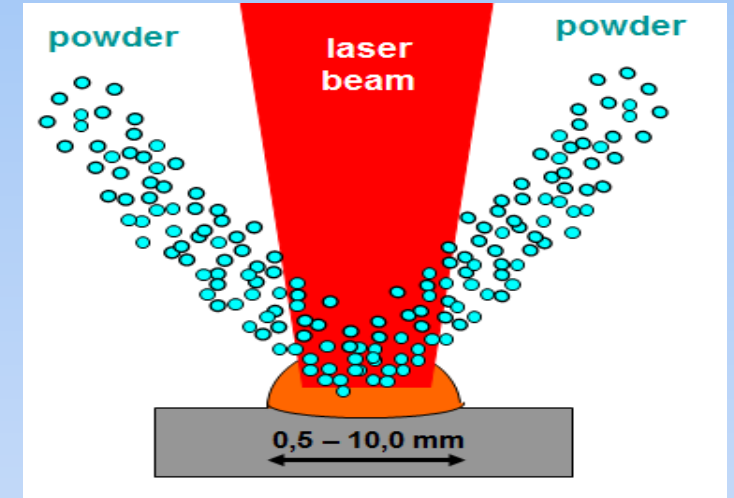
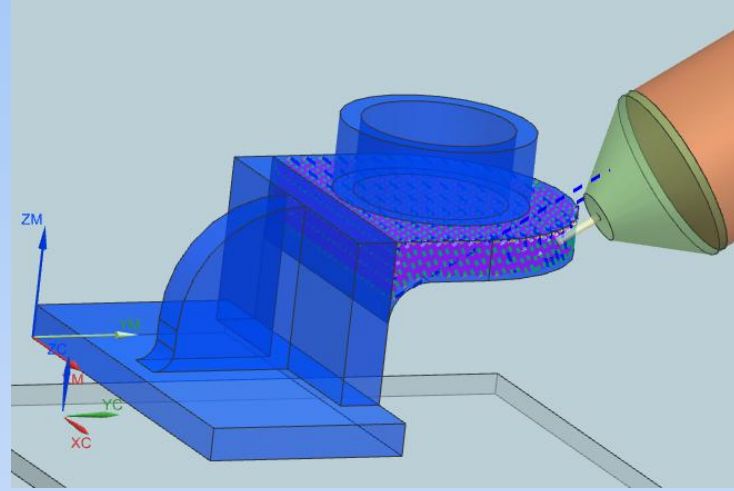
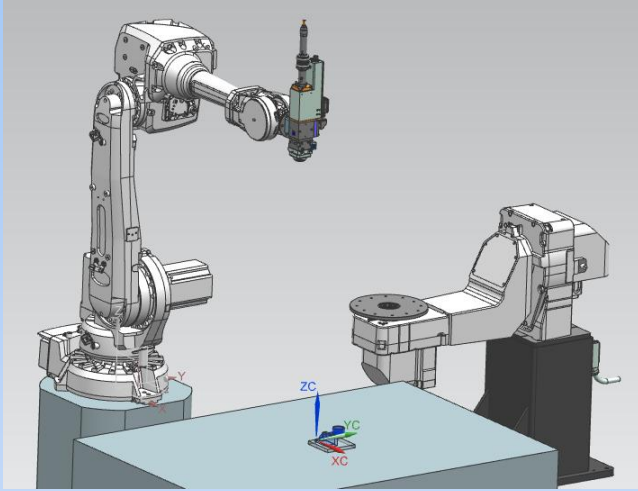


Summary

- **What is a DED system**
- **How it Works**
 - The 6 axis robot arm **ABB IRB 4600**
 - The Laser head **(3-6)KW**
 - The Powder feeder **(5-50) g/min**
- **First attempt to powder detection** (August - September 2019 Erasmus+)
 - Inductance meter
 - Frequency meter
 - Superheterodyne stabilized frequency meter (first attempt)
- **Second attempt to powder detection** (February - April 2020 Internship)
 - State of art of powder flow measurement systems
 - Light absorption/scattering method
 - Super heterodyne method (second attempt)
 - Beat (acoustics) method
- **Closed Loop Control**
 - Arduino MKR 1010 like control system

What is a DED system

Laser cladding



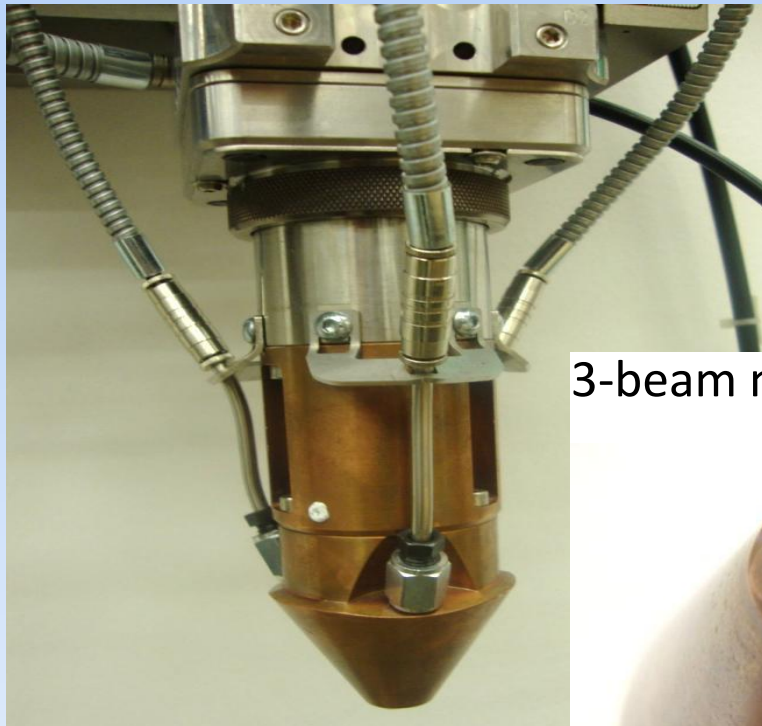
Robot arm ABB IRB 4600



Laser head

Principles

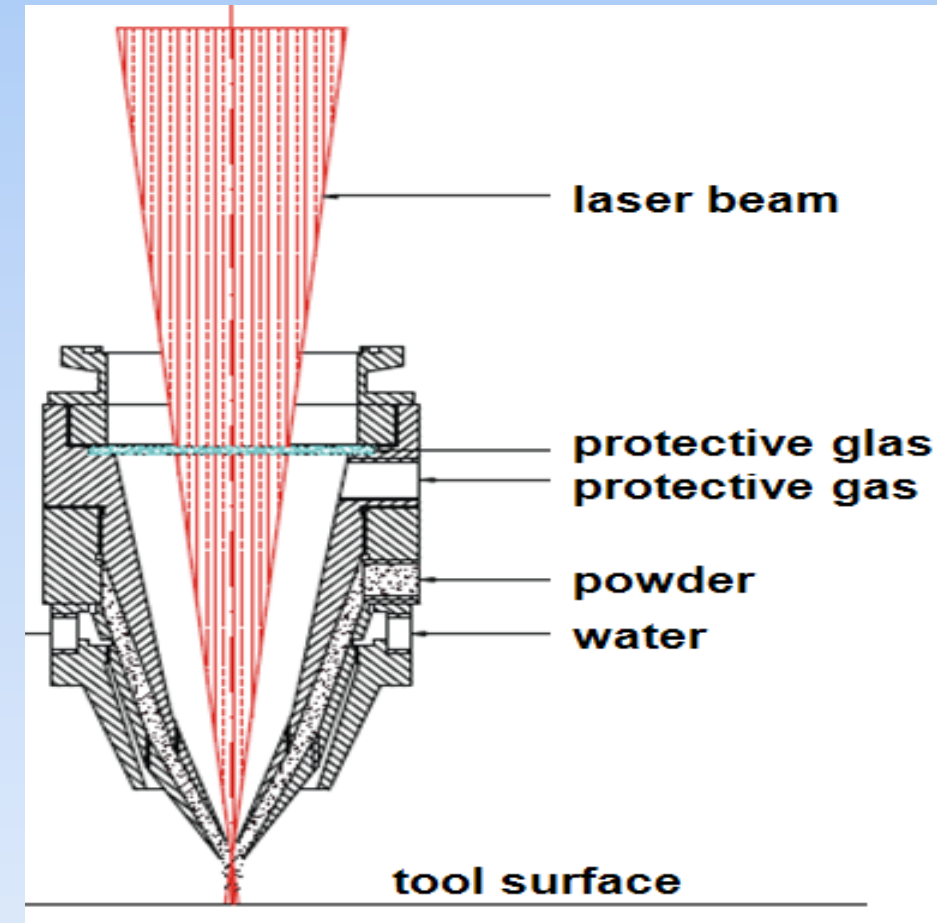
- Coaxial nozzle
 - homogeneous powder flow



3-beam nozzle



splitter

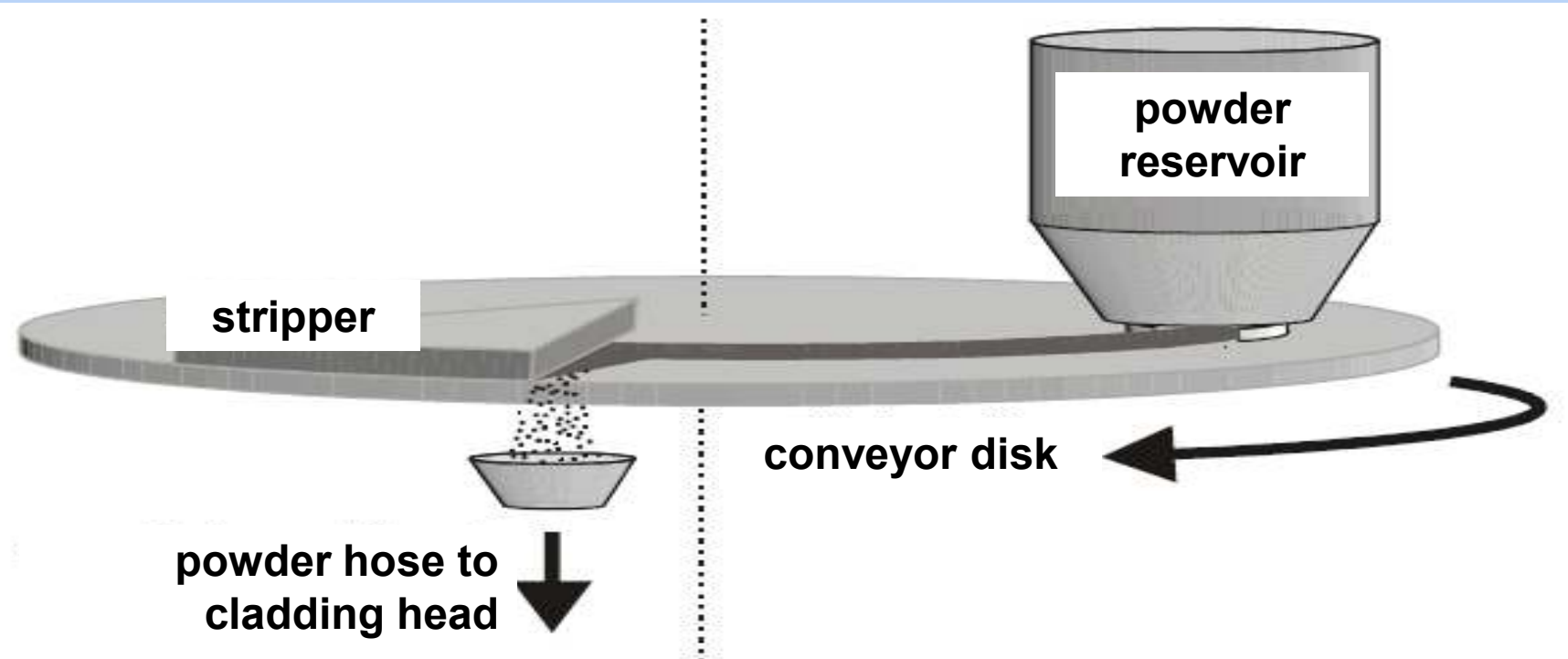


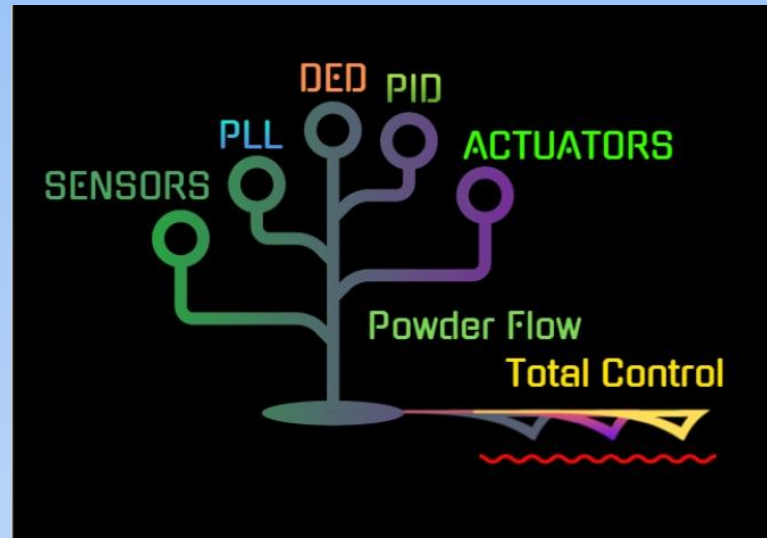
Powder feeder

Principles

■ Powder feeding

- powder feed rate controlled via rotation speed of conveyor disk





Powder Flow Total Control

Powder flow control of L-DED based additive manufacturing for zero defect metal components

First attempt to powder detection (August - September 2019 Erasmus+)



In August 2019 I was in charge of studying the possible principles for detecting and measure the flow of dust through the machine.

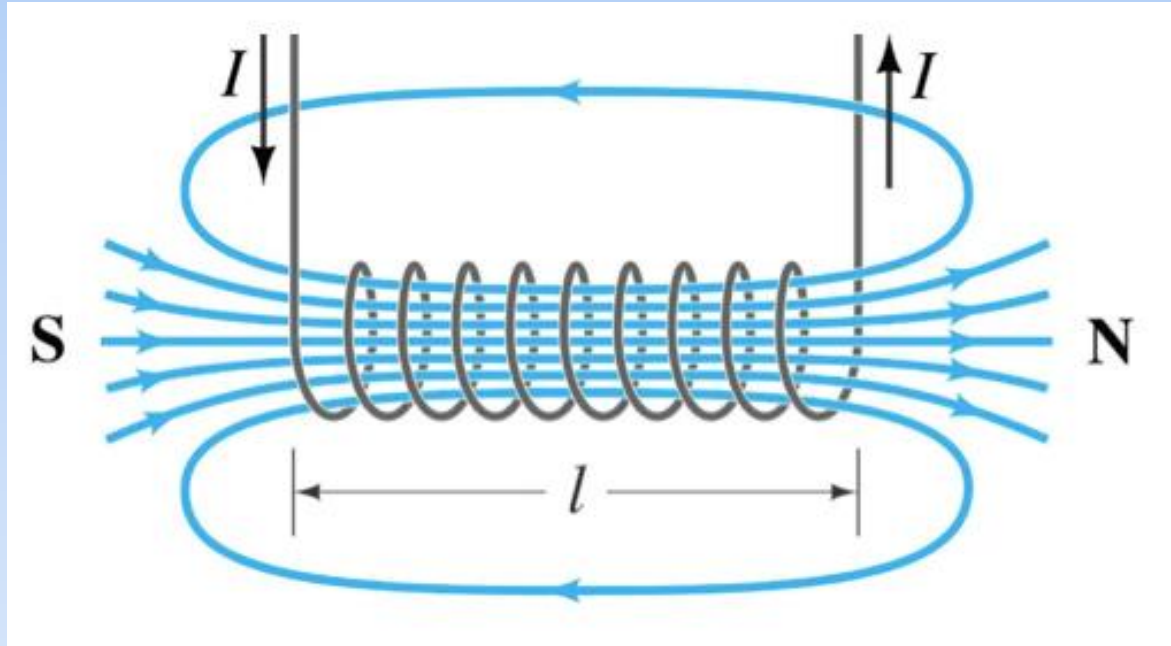
Input data:

- Powder flow rate between 2.5– 50 g/min
- Granulometry powder between 45 – 105 μ m
- Argon gas flow at 5 l/min
- Almost always the material is metallic

Firsts attempts to powder detection (August - September 2019 Erasmus+)



Because of the nature of the powder (metallic) my first idea was to try to detect it with one inductive method:



$$L = \mu \frac{N^2 S}{\ell}$$

First attempt to powder detection (August - September 2019 Erasmus+)



The solenoid formula shows that the self-induction coefficient L of the solenoid is proportional to the following physical quantities:

- μ : magnetic permeability of the solenoid core [H/m]
- N : number of turns
- S : solenoid cross-section area [m]
- l : solenoid length [m]

$$L = \mu \frac{N^2 S}{l}$$

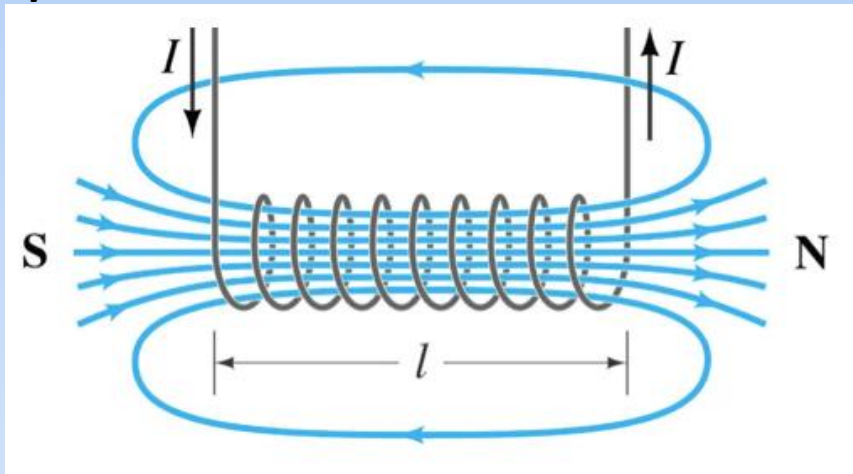
if the mechanical measurements of the solenoid are fixed, ($N S l = \text{cost}$) the variation of the self-induction coefficient L is proportional on the variation of the permeability of the core μ contained by the solenoid.

Firsts attempts to powder detection (August - September 2019 Erasmus+)



The μ coefficient, is the product of two other coefficients:

- μ_0 : **vacuum permeability**
- μ_r : **relative permeability**



Permeabilit  magnetica

$$\mu = \mu_0 \cdot \mu_r$$

Permeabilit  del vuoto

Permeabilit  relativa

$$\mu_0 = 4 \cdot \pi \cdot 10^{-7} \left[\frac{H}{m} \right]$$

*Because the vacuum permeability is constant, the variation of L is induced only if there is variation in the **relative permeability** μ_r*

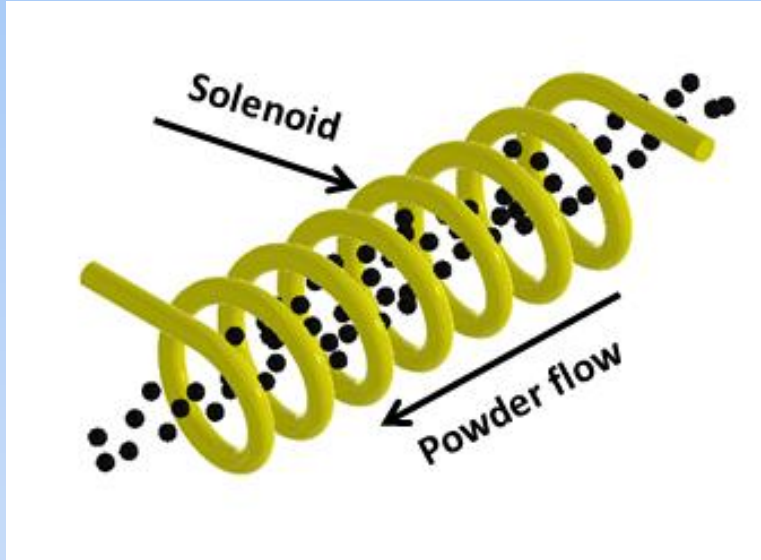
First attempt to powder detection (August - September 2019 Erasmus+)



The μ_r coefficient, it depends of the type of material as shown in the following table with some materials:

Sostanze	Permeabilità magnetica relativa μ_r		Sostanze	Permeabilità magnetica relativa μ_r
Diamagnetiche:			Paramagnetiche:	
Bismuto	0,999834		Aria	1,0000004
Oro	0,999964		Ossigeno	1,0000021
Mercurio	0,999968		Magnesio	1,000012
Argento	0,999974		Alluminio	1,000023
Diamante	0,999978		Tungsteno	1,000068
Piombo	0,999983		Titanio	1,000071
Rame	0,9999902		Platino	1,000360
Acqua	0,9999912		Uranio	1,000400
Sodio	0,9999976		Ferromagnetiche:	
Idrogeno	0,99999999901		Ferro	5.500
Azoto	0,99999999950		Permalloy	25.000
Anidride carbonica	0,99999999977		Mumetal	100.000
Vuoto:	1			

First attempt to powder detection (August - September 2019 Erasmus+)



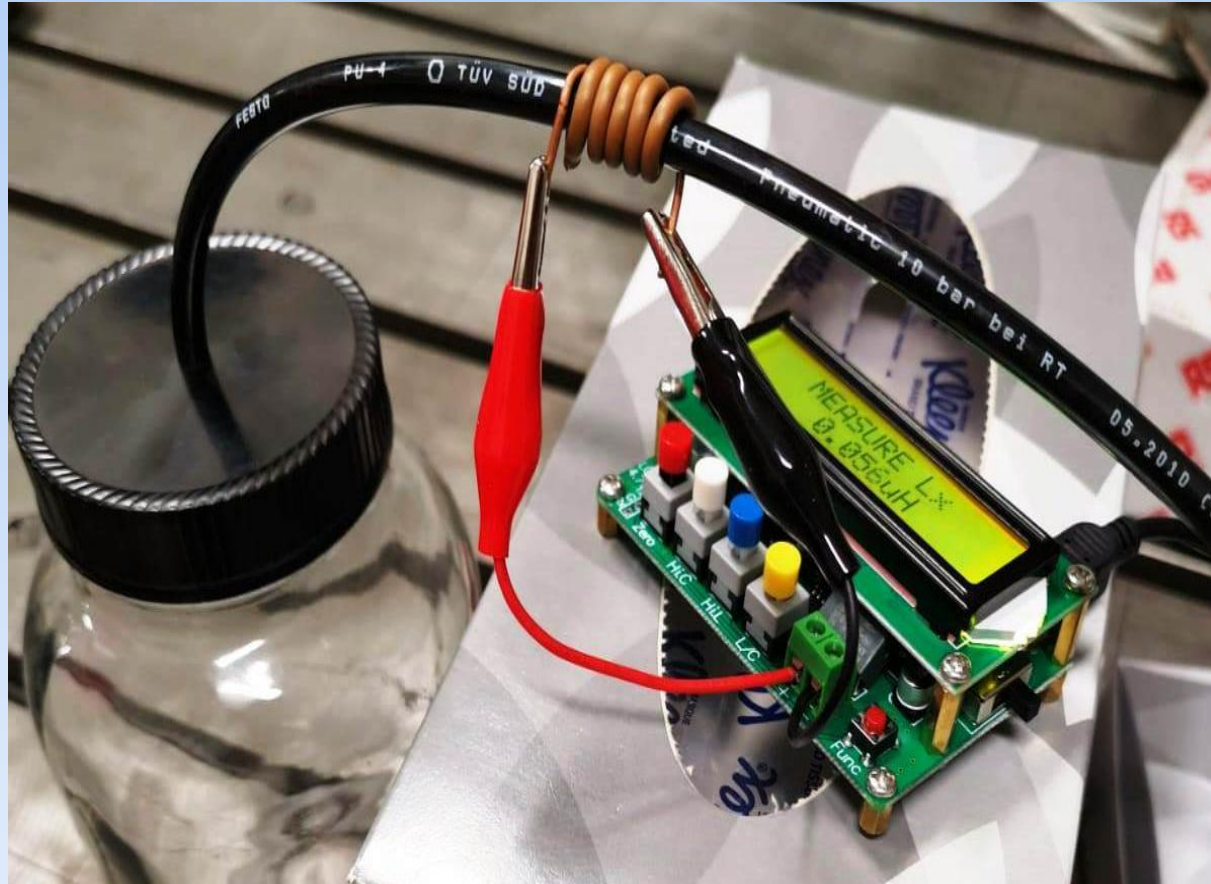
- If we imagine that the core inside the solenoid is made up of a heterogeneous mixture of metal dust and gas e.g. argon, it is theoretically possible to obtain a variation of the self-induction coefficient L because the total magnetic permeability coefficient of the volume contained inside the solenoid varies.

*On the basis of this fact it can be demonstrated that each variation in the mass flow of dust contained within the solenoid “**variable core**”, corresponds to a variation in the self-induction coefficient L .*

First attempt to powder detection (August - September 2019 Erasmus+)



- In other words, if we are able to measure the variations of the auto-induction coefficient L of the solenoid, we can obtain a measure of the mass flow rate of the powder.



- For this reason we have try to measure the variation in L with a simple commercial L/C meter.
- After some proves we have find out that the variation was to low to be detected by commercial L/C meter because of the poor accuracy

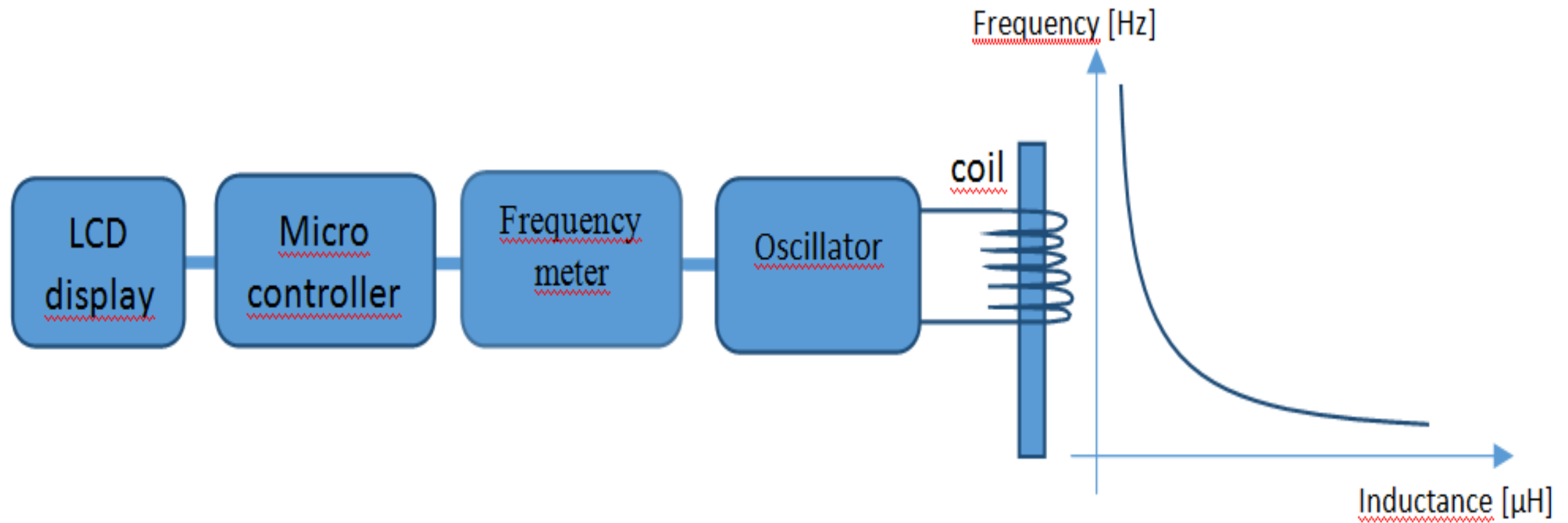
First attempt to powder detection (August - September 2019 Erasmus+)

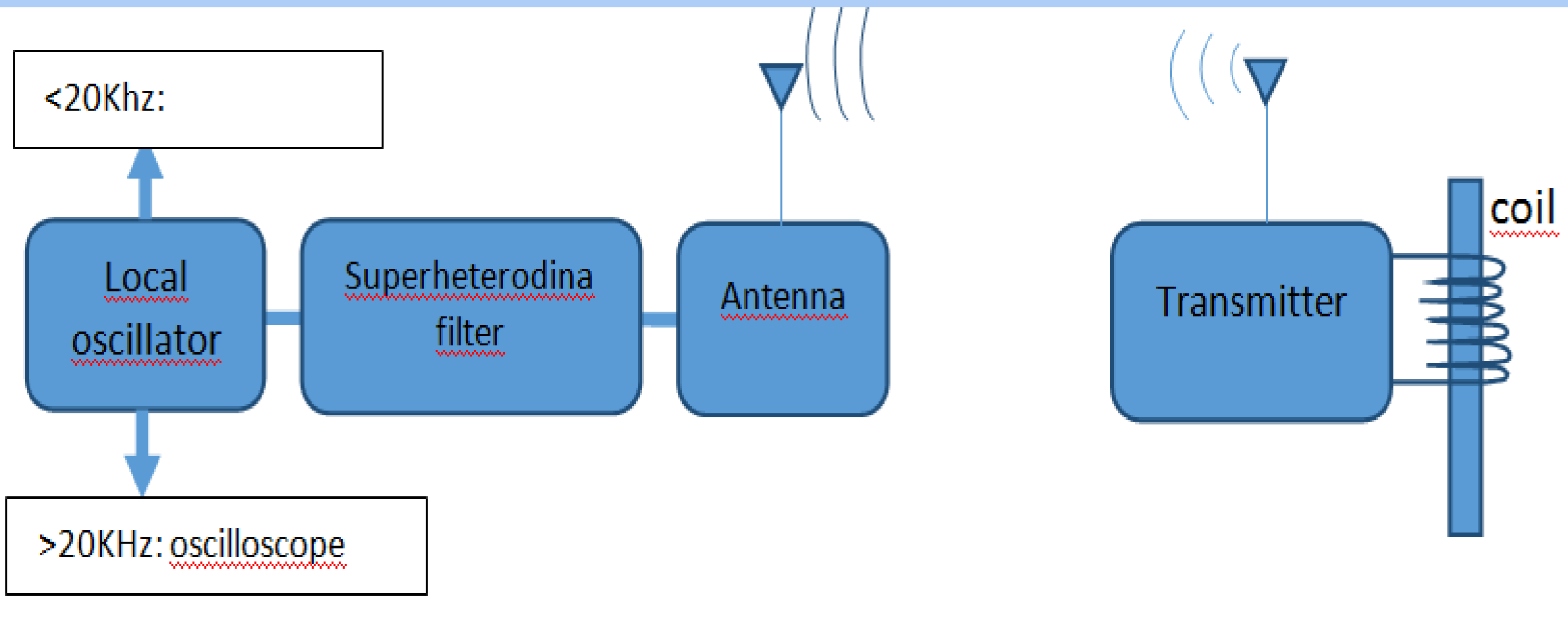


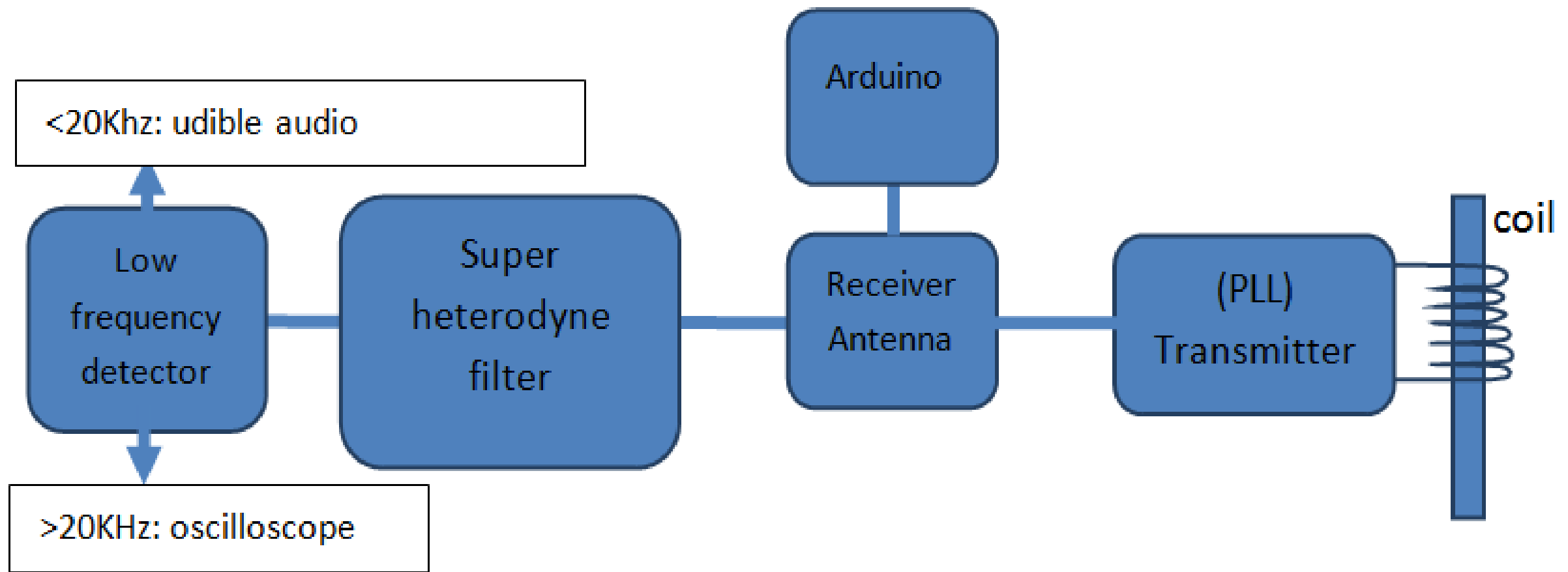
- Studying the schematic of the commercial instrument we have find out that the measure of the L was performed by a circuit that using the solenoid L and a capacitor of known value C, is able to measure the frequency of oscillation of the group L/C, in accordance with the following formula:

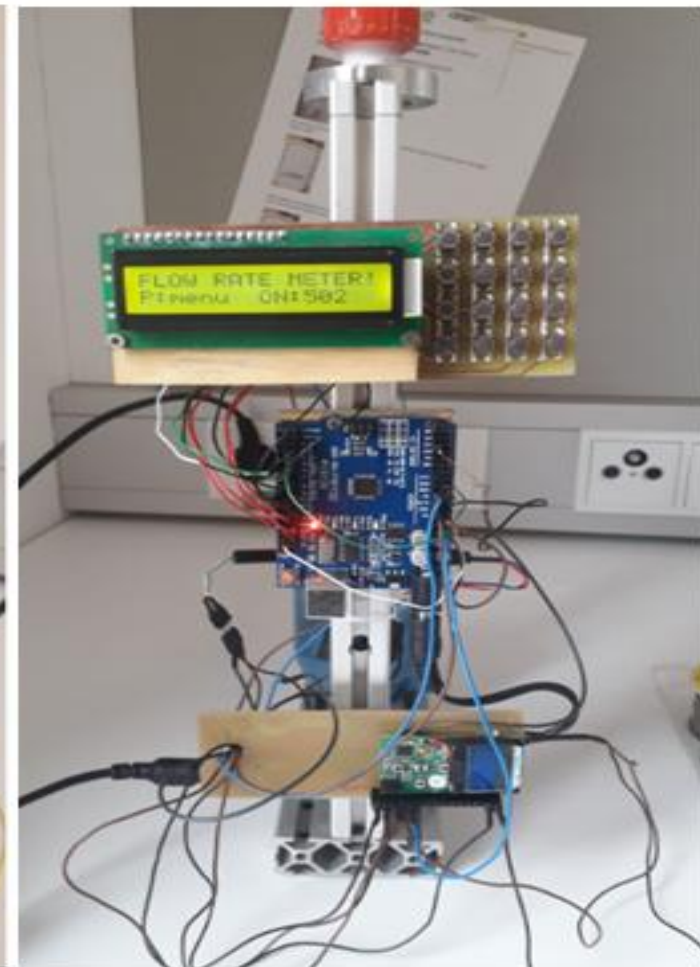
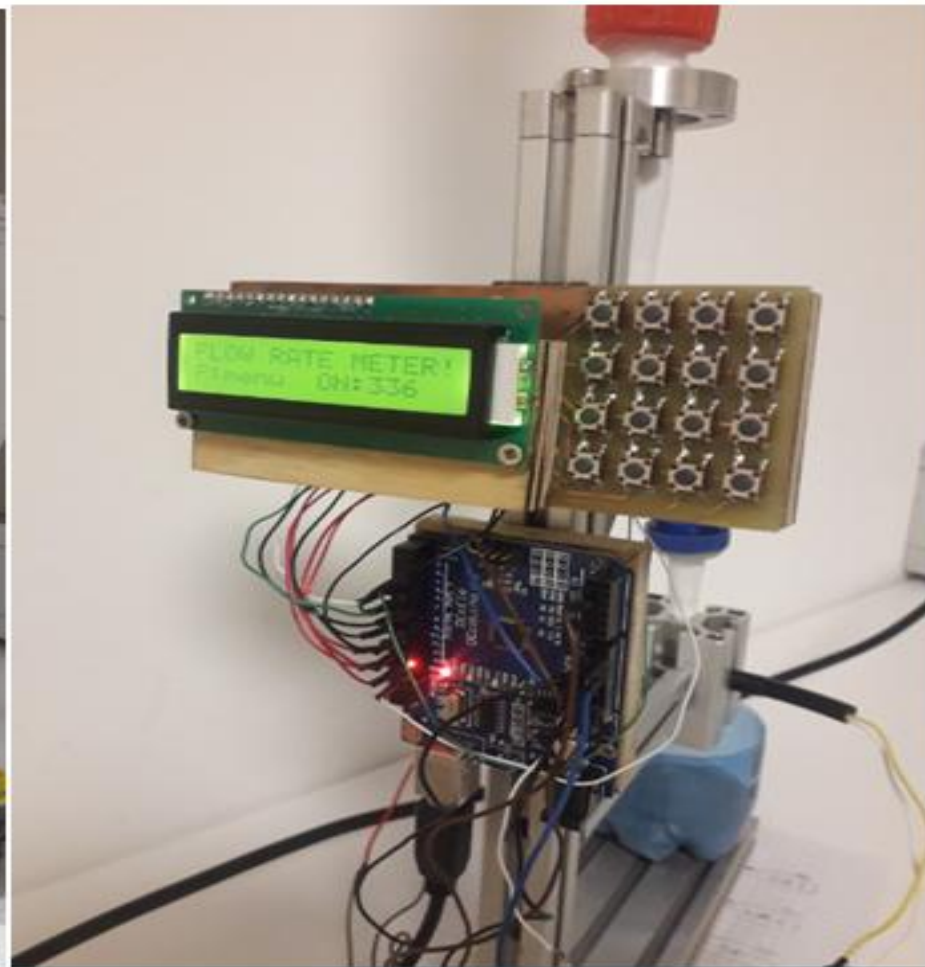
$$f_0 = \frac{1}{2\pi\sqrt{L_x C}} \longrightarrow L_x = \frac{1}{C(2\pi f_0)^2}$$

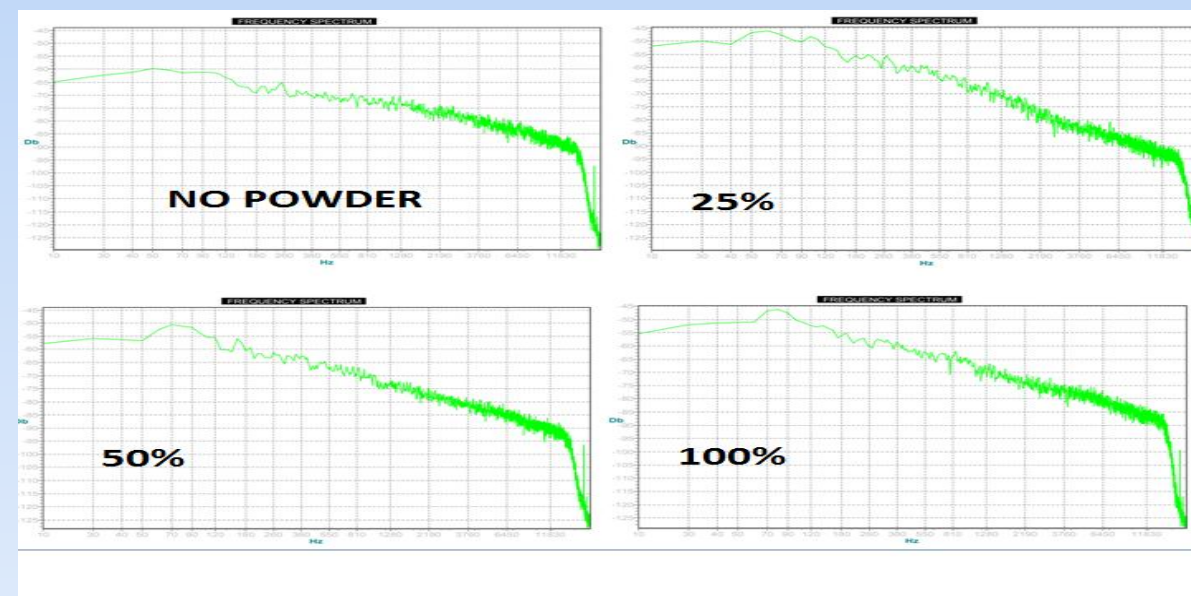
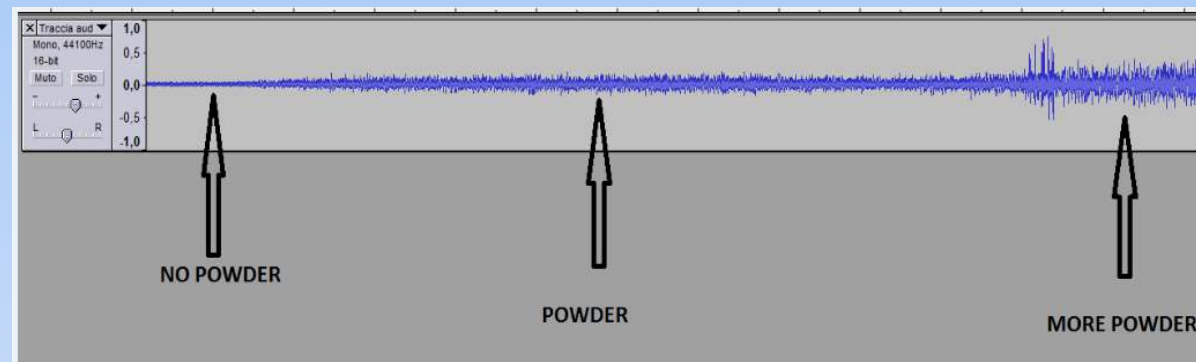
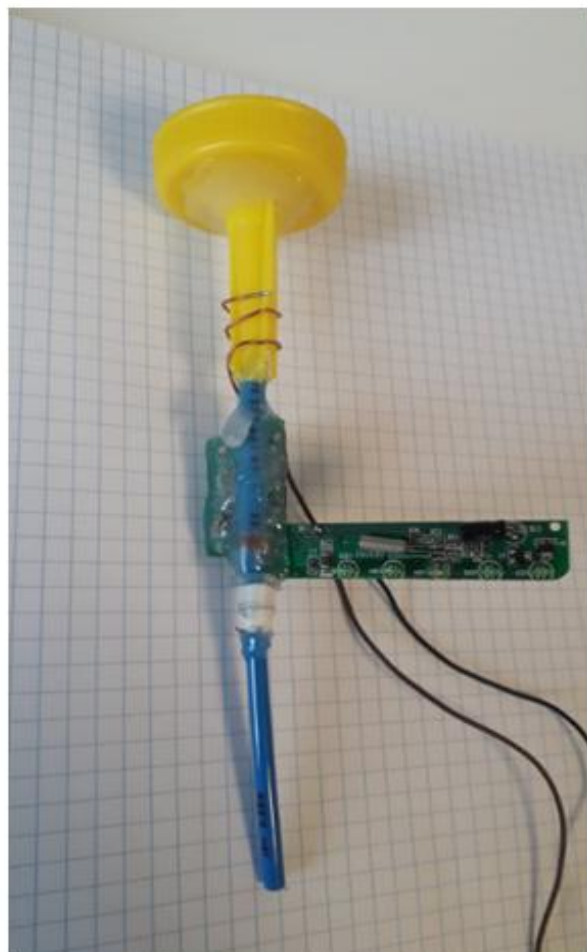
In other words if we can measure the frequency of the group L_x/C we can estimate the variation of the L_x

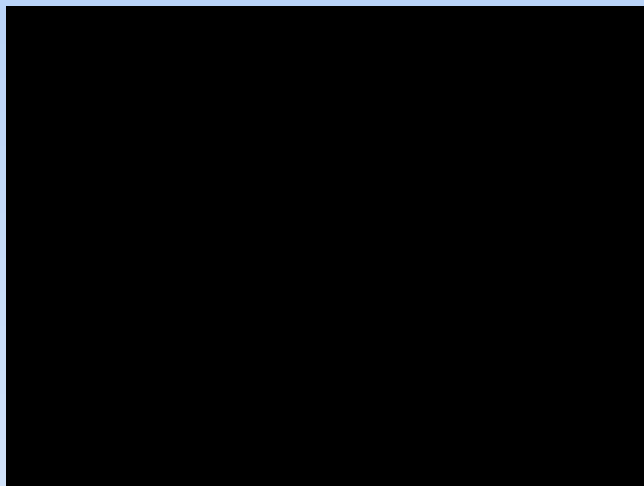


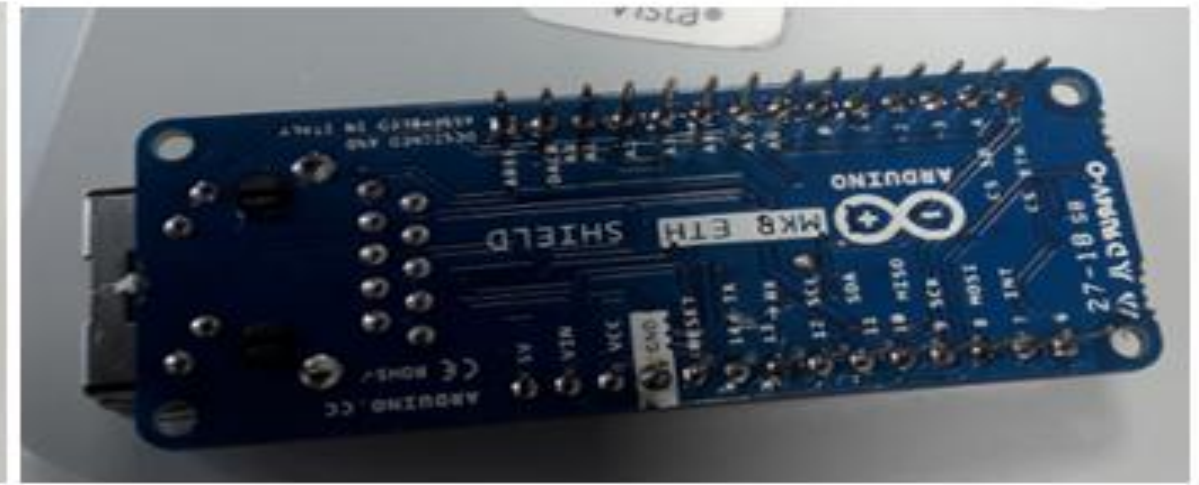
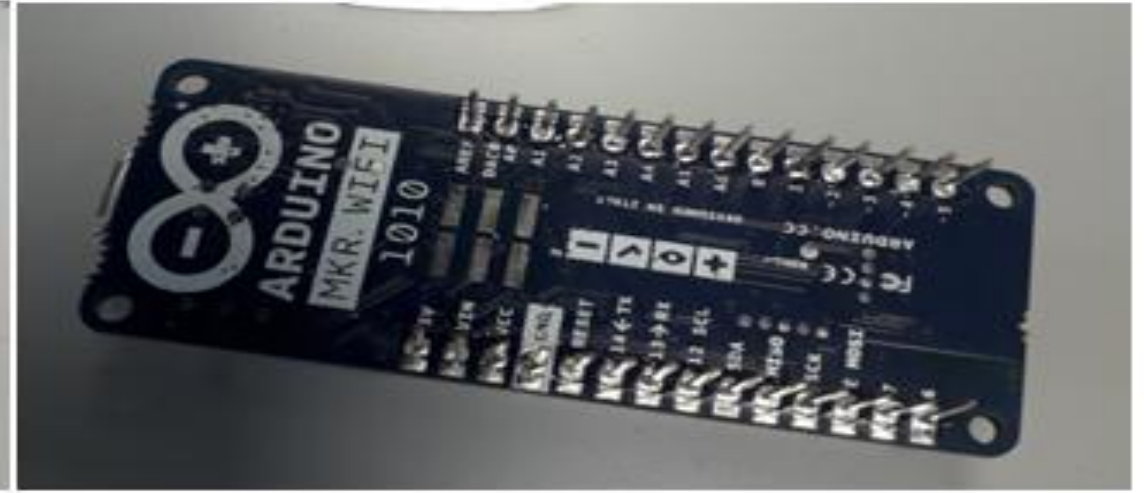


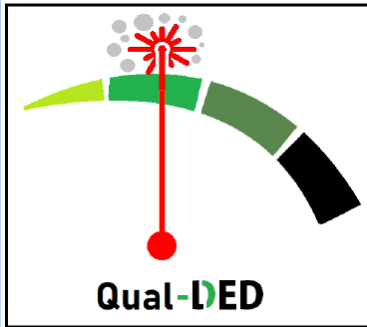












JOANNEUM
RESEARCH



FFG
Forschung wirkt.

This work has been done in the framework of the QuaL-DED: Total control of laser-based additive manufacturing for zero defect metal components. This project is funded by the Austrian Research Funding Agency (FFG), under the grant number 877386, within the 32. Call of the Factories of the Future (Produktion der Zukunft 32 AS).